

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The UDP protocol reveals that: The log shows that source IP 192.51.100.15 attempted to make queries to the DNS server at address 203.0.113.2 via port 53 using the UDP protocol.

This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message: The response traffic received uses the ICMP protocol to report a delivery failure.

The port noted in the error message is used for: The server repeatedly responded with the message "udp port 53 unreachable".

The most likely issue is: This means that the client's device attempted to translate the website name yummyrecipesforme.com into an IP address, but the DNS server was unable to process the request because the port responsible for that task (port 53) was unavailable.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

Time incident occurred: 13:24:32.192571

Explain how the IT team became aware of the incident: Several clients reported being unable to access the website www.yummyrecipesforme.com, receiving a "destination port unreachable" error after a prolonged timeout.

Explain the actions taken by the IT department to investigate the incident: Using the tcpdump tool, it was confirmed that outgoing requests were not receiving a response from the DNS service, but rather an ICMP error message indicating that port 53 was not listening for requests.

Note key findings of the IT department's investigation (i.e., details related to the port affected, DNS server, etc.): The website remains inaccessible to external users because their browsers cannot resolve the IP address needed to load the page.

Note a likely cause of the incident: El servicio DNS en el servidor **203.0.113.2** se ha detenido inesperadamente o existe una configuración reciente en el firewall que está bloqueando el tráfico **UDP** entrante en el puerto **53**.